

Supplementary Information for “Resistance-Aware Polypharmacology of Antimalarial Leads from African Natural Products: Docking, Network Scoring, and Targeted Molecular Dynamics”

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Table S1: Docking protocol validation: enrichment metrics and re-docking accuracy. Values reproduce the corrected benchmarks from the parent study (V2-corrected grids; AutoDock Vina + DiffDock consensus). MMV Malaria Box values are consensus-score ROC-AUCs for the three targets with MMV docking data (PfDHFR, PfCRT, PfATP4); DEKOIS 2.0 validation used AutoDock Vina only for PfDHFR. BEDROC values use $\alpha = 20$ (MMV). Re-docking RMSD is the heavy-atom RMSD between the top-scoring docked pose and the co-crystallised ligand; MTX redocking failed (RMSD ≈ 30 Å).

Target	Method	ROC-AUC	EF1 %	EF5 %	BEDROC	RMSD (Å)
PfDHFR (7F3Y)	Vina redock	–	–	–	–	N/A
	MMV consensus (Vina+DiffDock)	0.924	–	2.57	1.309	–
	DEKOIS (Vina only)	0.450	–	0.00	–	–
PfCRT (6UKJ)	MMV consensus (Vina+DiffDock)	0.971	–	1.11	9.434	–
PfATP4 (9N10)	MMV consensus (Vina+DiffDock)	1.000	–	2.00	1.810	–
PfClpR (4GM2; not PfClpP)	MMV enrichment	not evaluated	–	–	–	–

Identity note. PDB 4GM2 is PfClpR rather than PfClpP; it is not used as PfClpP validation.

Table S2: Endpoint-analysis interpretation for the four parent-study consensus complexes used for production MD. These systems are distinct from the 17 set-C polypharmacology candidates. Mean $\pm$ standard deviation is shown only for the single interpretable MM-GBSA result from 101 snapshots; dissociated or conversion-corrupted systems are reported as N/A. Mutant RRS is a docking-based analysis of set C and is not assigned to these MD systems.

Ligand	Target	ΔG_{bind}	Std. dev.	Interpretation
164	PfClpR-labelled cohort*-WT	–	–	N/A; ligand dissociated [‡]
201	PfDHFR-WT	–	–	N/A; ligand dissociated [‡]
214	PfCRT-WT	–18.25	0.40	Interpretable (gmx_MMPBSA)
438	PfATP4-WT	–	–	N/A; conversion corrupted [§]

*PDB 4GM2 is PfClpR rather than PfClpP; the 164-labelled system is therefore not structural validation of PfClpP.

[‡]Ligand dissociated during production; no binding free energy is inferred.

[§]Excluded due to two-chain topology incompatibility with CHARMM36-to-AMBER conversion.

Scope of the MD evidence. The parent-study trajectories provide a targeted structural check on four wild-type complexes; they do not constitute candidate-specific Set-C production MD. A 16-system Set-C MD pilot (PP-01 and PP-02 against the six-mutant panel, 10 ns each) passed the

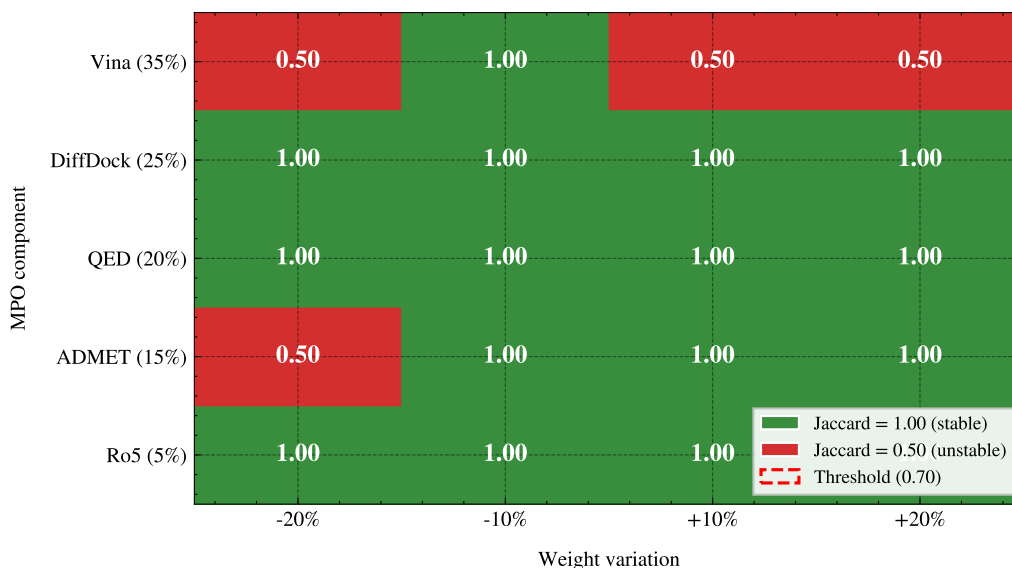


Figure S1: MPO sensitivity analysis heatmap. Each cell shows the Jaccard similarity of the top-20 hit list when the corresponding MPO weight is varied by $\pm 20\%$. Green (1.00) indicates the hit list is unchanged; red (0.50) indicates substantial reordering.

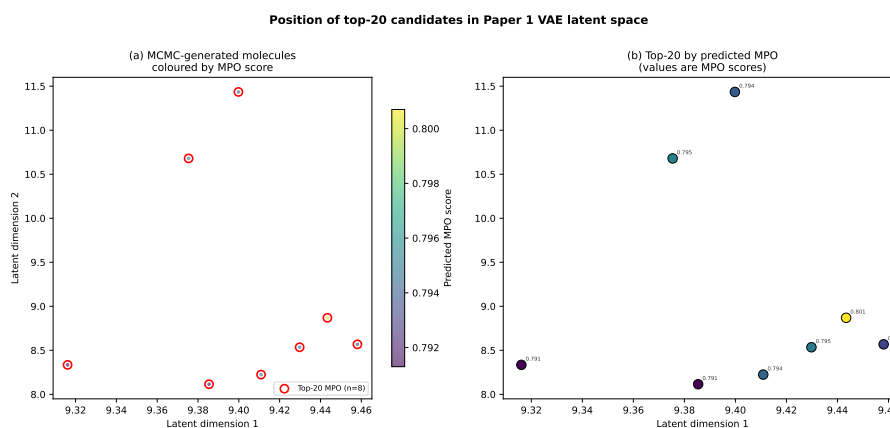


Figure S2: Position of the top-20 candidates in the Paper 1 VAE latent space. Molecules are coloured by predicted MPO score; top-20 by MPO are highlighted in red.

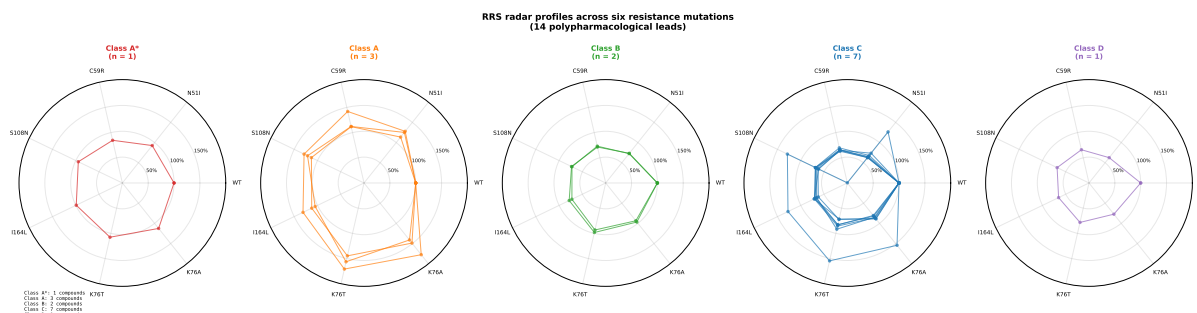


Figure S3: Per-compound, docking-derived RRS profiles across all six resistance mutations for the 17 set-C polypharmacological leads. The four parent-study MD systems are not included. Each axis represents one mutation; concentric rings indicate 50 %, 100 %, and 150 % RRS (WT-normalised). Larger radial values indicate greater preservation of the docking score relative to the corresponding wild-type target; the plot does not encode absolute binding potency.

Table S3: Predicted ADMET profile of the 17 set-C polypharmacology candidates (ADMET-AI, parent-library screening). All values are machine-learning predictions; experimental ADMET profiling and three-tool concordance were outside the scope of this study. Range across candidates: LogS [0.34, 0.73], CYP3A4 inhibition ≥ 0.5 for 7/17 candidates, Caco-2 permeability ≥ 0.5 for 15/17, Clint_h [0.48, 0.63].

Compound	ADMET-AI LogS	CYP3A4 prob.	Caco-2 prob.	Clint _h
PP-01	0.34	0.39	0.63	0.60
PP-02	0.46	0.97	0.86	0.55
PP-03	0.43	0.62	0.95	0.55
PP-04	0.40	0.58	0.92	0.63
PP-05	0.61	1.00	0.35	0.51
PP-06	0.54	0.90	0.97	0.60
PP-07	0.36	0.02	0.75	0.62
PP-08	0.59	1.00	0.41	0.56
PP-09	0.65	0.13	0.81	0.57
PP-10	0.49	0.13	0.93	0.62
PP-11	0.39	0.06	0.96	0.58
PP-12	0.49	0.00	0.98	0.48
PP-13	0.57	0.97	0.98	0.53
PP-14	0.66	0.05	0.68	0.49
PP-15	0.66	0.49	0.95	0.51
PP-16	0.54	0.25	0.91	0.50
PP-17	0.73	0.01	0.82	0.50

Table S4: ACSI weight sensitivity analysis. Each of the four ACSI weights (w_{DrugBank} , w_{ANPDB} , w_{fsp^3} , w_{NPL}) was varied by $\pm 20\%$ while keeping the remaining weights proportional. The Jaccard index measures overlap of the resulting top-20 ACSI-ranked list with the baseline ranking. A Jaccard ≥ 0.70 (14/20 compounds stable) indicates robustness.

Weight varied	Variation (%)	Mean ACSI (top-20)	Std.dev.	Jaccard vs. baseline
<i>Data to be populated after ACSI computation.</i>				

prespecified trajectory-QC gate (16/16) and is reported as a secondary analysis; its trajectory-derived MD-RRS and mutant-state MM-GBSA estimates are summarised in table S7 and compared with docking RRS in fig. S4.

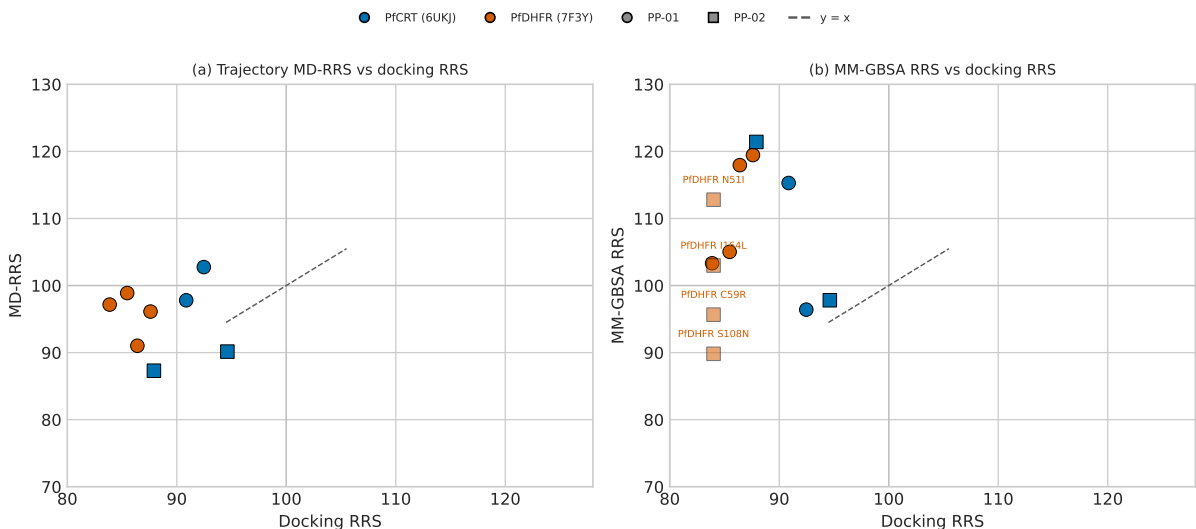


Figure S4: Set-C MD pilot trajectory-derived RRS estimates versus docking RRS (WT-normalised, WT = 100). Panel (a): MD-RRS (mean-minimum-distance geometry) versus docking RRS for the eight mutant systems with both estimates; panel (b): MM-GBSA RRS versus docking RRS, with the four PP-02 PfDHFR systems (no docking WT reference in the pilot) shown on the MM-GBSA axis only. Points are coloured by target (PfCRT, blue; PfDHFR, orange) and shaped by candidate (PP-01, circle; PP-02, square). Docking predicted looser mutant binding for all systems (RRS < 100); the trajectory-derived estimates do not reproduce this pattern. Secondary, exploratory analysis; no primary claim is based on it.

The stable identifiers were retrieved from PlasmoDB/VEuPathDB annotations on 12 August 2026 (PlasmoDB Consortium, 2026); record-level permalinks and the retrieval date are preserved in the machine-readable CSV. The exact database release/build was not captured in the source snapshot and is therefore not treated as an inferred annotation version. Because the present study does not provide an independent multi-gene targetome or a prespecified parasite-proteome background, no GO or pathway-enrichment test was performed. This table therefore supplies biological identity and mutation provenance without converting docking scores into pathway-level evidence (Trapotsi et al., 2022).

References

- PlasmoDB Consortium. PlasmoDB: The *Plasmodium* genomic resource. <https://plasmodb.org/>, 2026.
- Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.

Use of Artificial Intelligence

During the preparation of this work the authors used AI assistants to support code development and data analysis scripting. The authors reviewed and edited all AI-assisted output and take

Table S5: Stable *Plasmodium falciparum* target identifiers and mutation context used in this study. Identifiers are reported for annotation and provenance only; they do not constitute pathway-level target engagement, causal network evidence, or experimental validation. The accompanying source-data record preserves the target annotations and retrieval provenance.

Target	Stable gene ID	Structural context	Mutation or evidence boundary
PfDHFR	PF3D7_0417200	7F3Y WT template	N51I, C59R, S108N, I164L; antifolate-resistance states
PfCRT	PF3D7_0709000	6UKJ WT template	K76T clinical resistance marker; K76A comparator state
PfATP4	PF3D7_1211900	9N10 parent-study structure	Target annotation only; not part of the Set-C mutation panel
PfClpP	PF3D7_0307400	Proteolytic Clp subunit	Not assigned to PDB 4GM2; no PfClpP claim is inferred from the 4GM2 record
PfClpR	PF3D7_1436800	4GM2, PfClpR(P61–E244)	4GM2 is PfClpR, not PfClpP; parent-study identity retained explicitly

Table S6: Cohort and estimand separation. The Set-C docking cohort and the parent-study MD cohort are non-overlapping. The Set-C MD pilot passed the prespecified trajectory-QC gate (16/16) and is reported as a secondary, exploratory analysis whose MD-RRS and MM-GBSA estimates are not merged into the primary docking-RRS classification.

Evidence layer	Cohort	Systems	Primary analyses	Interpretation boundary
Set-C docking	PP-01–PP-17	136	WT/mutant docking; RRS, ACSI, PNS	Docking-derived prioritisation; not measured binding or MD-RRS
Parent-study MD	Ligands 164, 201, 214, 438	4 WT complexes × 10 ns	RMSD, contacts, bound-state assessment, MM-GBSA where valid	Separate structural check; one interpretable MM-GBSA estimate (PfCRT–214)
Set-C MD pilot	PP-01/PP-02	16 systems (16/16 QC PASS)	Trajectory QC, discriminative MD-RRS, MM-GBSA	Reported as secondary; not merged into primary RRS classification
Full Set-C MD-RRS	PP-01–PP-17	136 planned systems	Full trajectory QC and MD-RRS	Not computed; requires complete PASS-QC cohort

Table S7: Set-C MD pilot endpoint estimates (secondary, exploratory analysis). All 16 systems passed the prespecified trajectory-QC gate (16/16); ΔG_{bind} is the mean over 100 snapshots $\pm$ standard deviation. MMG-RRS and MD-RRS use the same mutant/WT ratio convention as the docking RRS (WT = 100); n/a = no WT reference available in the pilot for PP-02 PfDHFR (docking RRS was computed against the parent 17-candidate cohort only). The two systems flagged with * failed the first MM-GBSA pass (BOND/UB overflow from a protein split across the periodic boundary) and were re-run on PBC-whole trajectories.

Candidate	Target	State	ΔG_{bind} (kcal mol ⁻¹)	SD	MMG-RRS	MD-RRS	Docking RRS
PP-01	PfCRT	K76A	-35.29	1.17	115.3	97.8	90.86
PP-01	PfCRT	K76T	-29.51	2.65	96.4	102.8	92.47
PP-01	PfCRT	WT	-30.61	1.65	100.0	100.0	-
PP-01	PfDHFR	C59R	-26.56	1.91	105.0	98.9	85.47
PP-01	PfDHFR	I164L	-26.13	2.34	103.3	97.2	83.87
PP-01	PfDHFR	N51I	-29.83	1.50	118.0	91.0	86.40
PP-01	PfDHFR	S108N	-30.21	3.62	119.5	96.1	87.60
PP-01	PfDHFR	WT	-25.29	2.94	100.0	100.0	-
PP-02	PfCRT	K76A*	-24.53	1.27	97.8	90.2	94.62
PP-02	PfCRT	K76T	-30.45	1.38	121.4	87.3	87.91
PP-02	PfCRT	WT	-25.08	2.56	100.0	100.0	-
PP-02	PfDHFR	C59R	-28.85	1.88	95.7	102.0	n/a
PP-02	PfDHFR	I164L	-31.06	1.52	103.0	97.4	n/a
PP-02	PfDHFR	N51I	-34.02	2.68	112.8	72.2	n/a
PP-02	PfDHFR	S108N	-27.09	1.55	89.8	97.5	n/a
PP-02	PfDHFR	WT*	-30.16	2.06	100.0	100.0	-

full responsibility for the content of this article. No generative AI tool is listed as an author. All computational results, statistical analyses, and quantitative claims were generated and verified by the authors.